

CROWN INSTITUTE

FROM POSSIBILITY TO ACTUALITY

ICT313 Big Data for Software Development

LECTURE 10
Similarity

Similarity and Dissimilarity

- Similarity
  - Numerical measure of how alike two data objects are.
  - Is higher when objects are more alike.
  - Often falls in the range [0,1]
- Dissimilarity
  - Numerical measure of how different are two data objects
  - Lower when objects are more alike
  - Minimum dissimilarity is often 0
  - Upper limit varies
- Proximity refers to a similarity or dissimilarity

Similarity/Dissimilarity for Simple Attributes

p and *q* are the attribute values for two data objects.

Attribute Type	Dissimilarity	Similarity
Nominal	$d = \begin{cases} 0 & \text{if } p = q \\ 1 & \text{if } p \neq q \end{cases}$	$s = \begin{cases} 1 & \text{if } p = q \\ 0 & \text{if } p \neq q \end{cases}$
Ordinal	$d = \frac{\lvert p-q \rvert}{n-1}$ (values mapped to integers 0 to $n-1$, where n is the number of values)	$s = 1 - \frac{\lvert p-q \rvert}{n-1}$
Interval or Ratio	$d = \lvert p - q \rvert$	$s = -d$, $s = \frac{1}{1+d}$ or $s = 1 - \frac{d - min_d}{max_d - min_d}$

Table 5.1. Similarity and dissimilarity for simple attributes

Euclidean Distance

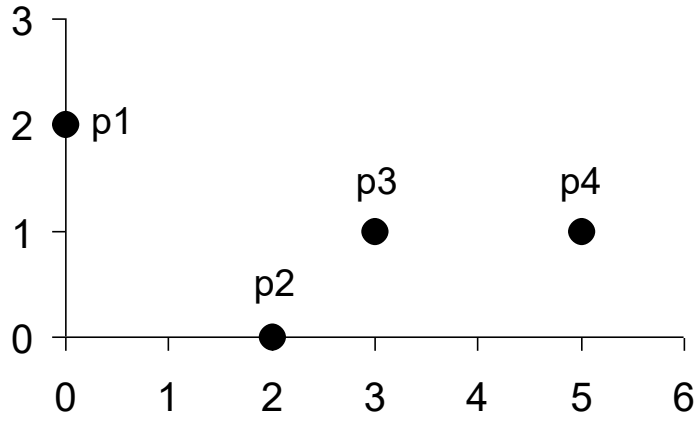
- Euclidean Distance

$$dist = \sqrt{\sum_{k=1}^{n} (p_k - q_k)^2}$$

  - Where n is the number of dimensions (attributes) and p_k and q_k are, respectively, the kth attributes (components) or data objects p and q.

- Standardization is necessary, if scales differ.

Euclidean Distance



point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

Distance Matrix

Minkowski Distance

- Minkowski Distance is a generalization of Euclidean Distance

$$dist = \left(\sum_{k=1}^{n} | p_k - q_k |^r \right)^{\frac{1}{r}}$$

  - Where *r* is a parameter, *n* is the number of dimensions (attributes) and p_k and q_k are, respectively, the kth attributes (components) or data objects *p* and *q*.

Minkowski Distance: Examples

- $r = 1$. City block (Manhattan, taxicab, L_1 norm) distance.
  - A common example of this is the Hamming distance, which is just the number of bits that are different between two binary vectors
- $r = 2$. Euclidean distance
- $r \rightarrow \infty$. "supremum" (L_{max} norm, L_∞ norm) distance.
  - This is the maximum difference between any component of the vectors
- Do not confuse *r* with *n*, i.e., all these distances are defined for all numbers of dimensions.

Minkowski Distance

Four users rated two movies as follows. We want to find the similarity of the users.
Why we may need this?

User	Movie 1	Movie 2
James	0	2
John	2	0
Sue	3	1
Anna	5	1

L1	James	John	Sue	Anna
James	0	4	4	6
John	4	0	2	4
Sue	4	2	0	2
Anna	6	4	2	0

L2	James	John	Sue	Anna
James	0	2.828	3.162	5.099
John	2.828	0	1.414	3.162
Sue	3.162	1.414	0	2
Anna	5.099	3.162	2	0

L_∞	James	John	Sue	Anna
James	0	2	3	5
John	2	0	1	3
Sue	3	1	0	2
Anna	5	3	2	0

Distance Matrix

Minkowski Distance

Four users rated two movies as follows. We want to find the similarity of the users.
Why we may need this?

User	Movie 1	Movie 2
James	0	2
John	2	0
Sue	3	1
Anna	5	1

Who is the most similar user to Anna?

Manhattan

L1	James	John	Sue	Anna
James	0	4	4	6
John	4	0	2	4
Sue	4	2	0	2
Anna	6	4	2	0

Euclidean

L2	James	John	Sue	Anna
James	0	2.828	3.162	5.099
John	2.828	0	1.414	3.162
Sue	3.162	1.414	0	2
Anna	5.099	3.162	2	0

Supremum

L_∞	James	John	Sue	Anna
James	0	2	3	5
John	2	0	1	3
Sue	3	1	0	2
Anna	5	3	2	0

Distance Matrix

Similarity Between Binary Vectors

- Common situation is that objects, *p* and *q*, have only binary attributes
- Compute similarities using the following quantities
  - M_{01} = the number of attributes where p was 0 and q was 1
  - M_{10} = the number of attributes where p was 1 and q was 0
  - M_{00} = the number of attributes where p was 0 and q was 0
  - M_{11} = the number of attributes where p was 1 and q was 1
- Simple Matching and Jaccard Coefficients

  SMC = number of matches / number of attributes

  $= (M_{11} + M_{00}) / (M_{01} + M_{10} + M_{11} + M_{00})$

  J = number of 11 matches / number of not-both-zero attributes values

  $= (M_{11}) / (M_{01} + M_{10} + M_{11})$

SMC versus Jaccard: Example

- *The following vectors show which items have been purchased by James and John:*

	Coke	Jam	Bread	Rice	Sugar	Milk	Butter	Cheese	Meat	Cake
James	1	0	0	0	0	0	0	0	0	0
John	0	0	0	0	0	0	1	0	0	1

- $M_{01} = 2$ (the number of attributes where p was 0 and q was 1)
- $M_{10} = 1$ (the number of attributes where p was 1 and q was 0)
- $M_{00} = 7$ (the number of attributes where p was 0 and q was 0)
- $M_{11} = 0$ (the number of attributes where p was 1 and q was 1)
-
- $SMC = (M_{11} + M_{00})/(M_{01} + M_{10} + M_{11} + M_{00}) = (0+7) / (2+1+0+7) = 0.7$
- $J = (M_{11}) / (M_{01} + M_{10} + M_{11}) = 0 / (2 + 1 + 0) = 0$

Cosine Similarity

- If d_1 and d_2 are two document vectors, then
- $$\cos(d_1, d_2) = (d_1 \bullet d_2) / ||d_1|| \, ||d_2|| ,$$
- where $\bullet$ indicates vector dot product and $||d||$ is the length of vector d.

- Example (word frequencies for two documents):

	Source	Destination	Wifi	Network	Connection	Program	Memory	Hardware	Test	IT
Document 1	3	2	0	5	0	0	0	2	0	0
Document 2	1	0	0	0	0	0	0	1	0	2

- $d_1 \bullet d_2 = 3*1 + 2*0 + 0*0 + 5*0 + 0*0 + 0*0 + 0*0 + 2*1 + 0*0 + 0*2 = 5$
- $||d_1|| = (3*3+2*2+0*0+5*5+0*0+0*0+0*0+2*2+0*0+0*0)^{0.5} = (42)^{0.5} = 6.481$
- $||d_2|| = (1*1+0*0+0*0+0*0+0*0+0*0+0*0+1*1+0*0+2*2)^{0.5} = (6)^{0.5} = 2.245$
- $\cos(d_1, d_2) = .3150$
- How do you interpret this?

Reference

- Tan, P., Steinbach, M., Karpatne, A., Kumar, V. (2019). Introduction to Data Mining, 2nd Edition, Pearson Education Limited.